

Conversion of Units Across Systems of Measure

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Measurement systems are standardized ways of expressing length, mass, capacity, and other quantities.

Two Common Systems

The two most common systems are the:

- ▶ **International System of Units (SI)**
- ▶ **U.S. Customary System**

Conversion Factors: Length

U.S. Customary	Metric
1 inch	2.54 cm
1 foot	30.48 cm
1 yard	0.9144 m
1 mile	1.609 km

Conversion Factors: Mass & Capacity

U.S. Customary	Metric
1 pound	0.454 kg
1 ounce	28.35 g
1 gallon	3.785 L
1 quart	0.946 L

How to Convert

To convert measurements, **multiply** by the appropriate conversion factor.

given quantity $\times$ conversion factor

= converted quantity

Examples

Example 1

Convert 6 inches to centimeters.

$$6 \times 2.54 = 15.24 \text{ cm}$$

Example 2

Convert 4 feet to meters.

$$4 \text{ ft} \times 30.48 \text{ cm} = 121.92 \text{ cm} = 1.2192 \text{ m}$$

Example 3

Convert 12 pounds to kilograms.

$$12 \times 0.454 = 5.448 \text{ kg}$$

Example 4

Convert 2 gallons to liters.

$$2 \times 3.785 = 7.57 \text{ L}$$

Practice Exercises

Practice

Convert each quantity to the indicated unit.

Practice 1

12 inches to centimeters

Practice 2

5 feet to meters

Practice 3

8 yards to meters

Practice 4

10 miles to kilometers

Practice 5

15 pounds to kilograms

Practice 6

24 ounces to grams

Practice 7

3 gallons to liters

Practice 8

7 quarts to liters

Practice 9

36 inches to centimeters

Practice 10

5 gallons to liters

Activities

Activity

Convert each quantity to the indicated unit.

Activity 1

18 inches to centimeters

Activity 2

9 feet to meters

Activity 3

15 yards to meters

Activity 4

25 miles to kilometers

Activity 5

20 pounds to kilograms

Activity 6

40 ounces to grams

Activity 7

6 gallons to liters

Activity 8

10 quarts to liters

Activity 9

48 inches to centimeters

Activity 10

8 gallons to liters

Thank you!